

FRANK PORTER
2016-2017

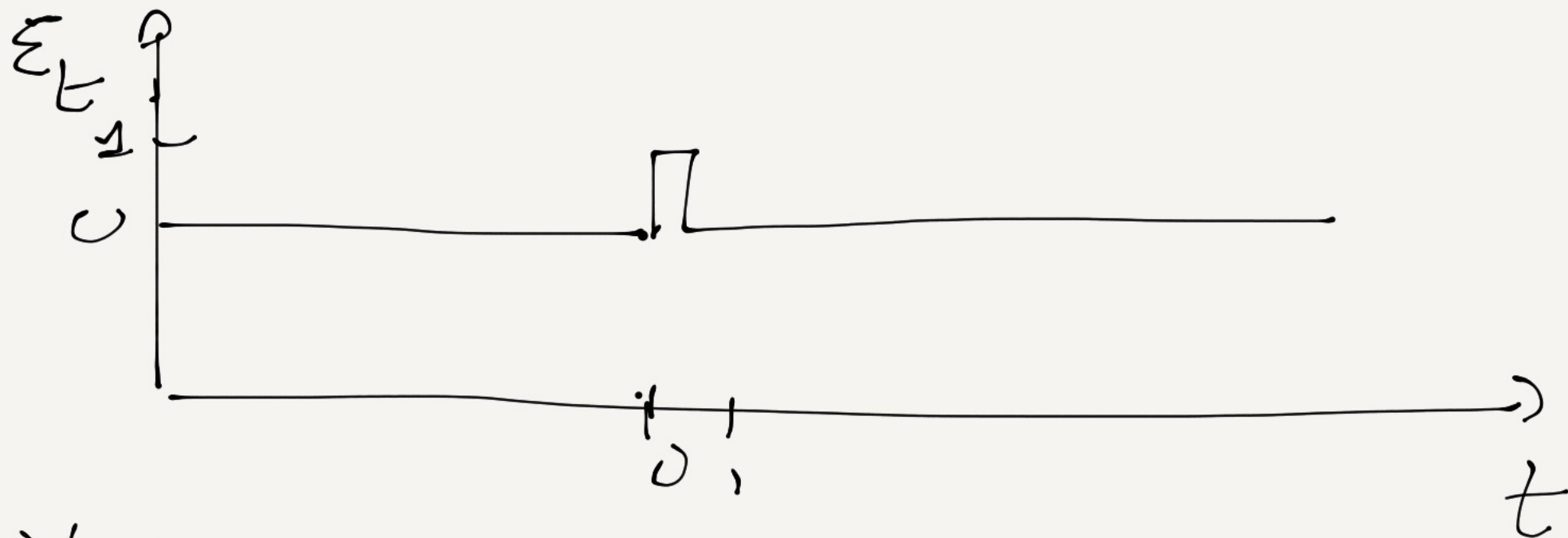
M1 MACRO

Group B

Session 16

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$X_t = \rho X_{t-1} + \varepsilon_t$$



—: ρ is large
—: ρ is small

X_t

~ = persistent shock
~ = temporary shock

• If $X_t = \rho B X_{t-1} + \varepsilon_t$ $(|\rho| < 1)$

$$X_t = \rho B X_t + \varepsilon_t$$

$$X_t (1 - \rho B) = \varepsilon_t$$

$$X_t = \frac{\varepsilon_t}{1 - \rho B} = (1 + \rho B + (\rho B)^2 + (\rho B)^3 + \dots) \times \varepsilon_t$$

$\hookrightarrow \tilde{W}_t = \frac{1}{1 - \rho_w B} \varepsilon_{w,t}$

$$\text{If } X_t = \frac{1}{(1 - \rho B)} \varepsilon_t$$

$$X_t (1 - \rho B) = \varepsilon_t$$

$$X_t = \rho X_{t-1} + \varepsilon_t = \rho (\rho X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t$$

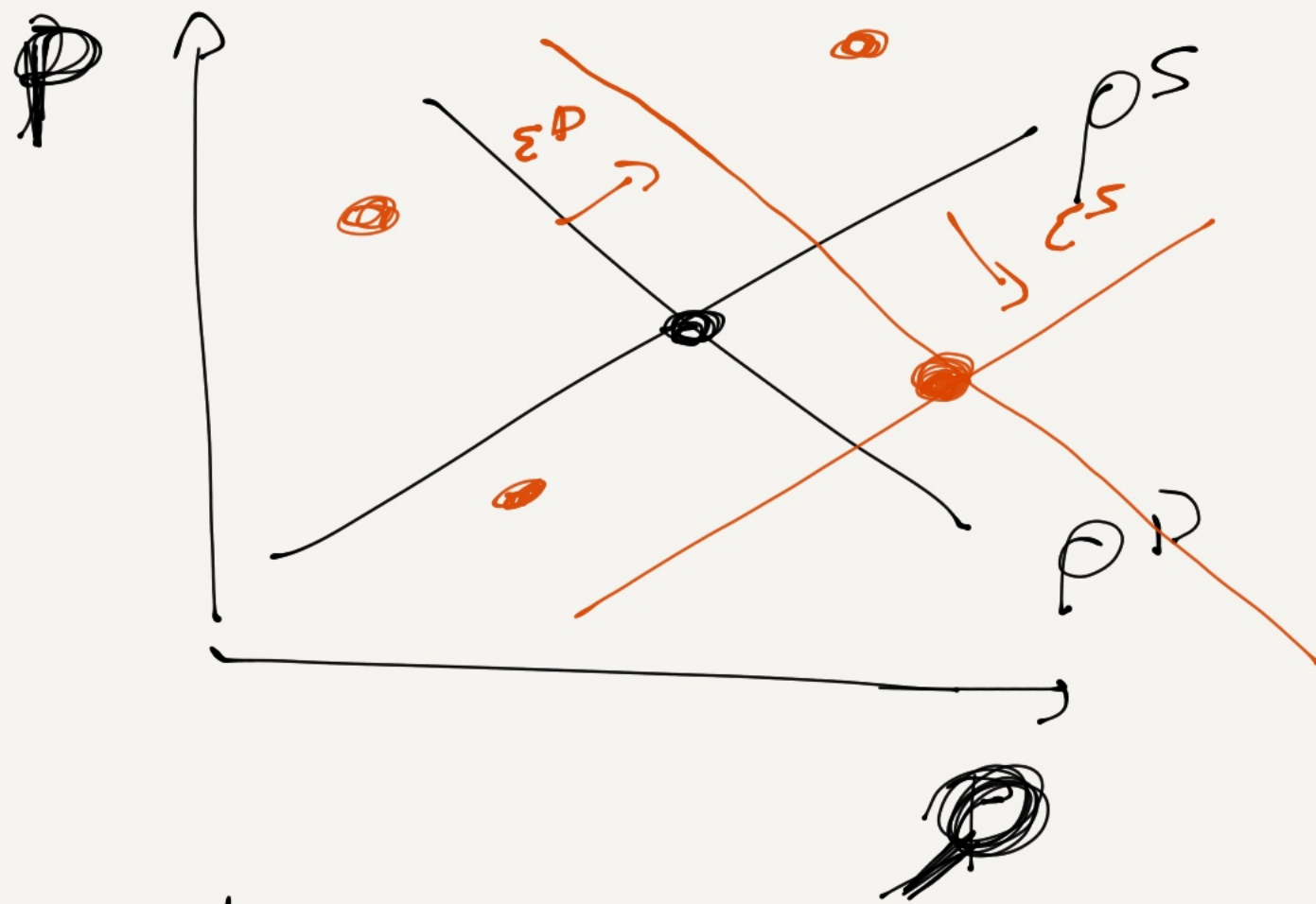
$$= \rho^2 X_{t-2} + \rho \varepsilon_{t-1} + \varepsilon_t$$

.....

$$= \varepsilon_t + \rho \varepsilon_{t-1} + \rho^2 \varepsilon_{t-2} + \rho^3 \varepsilon_{t-3} + \dots$$

$$= \varepsilon_t (1 + \rho B + \rho^2 B^2 + \rho^3 B^3 + \dots)$$

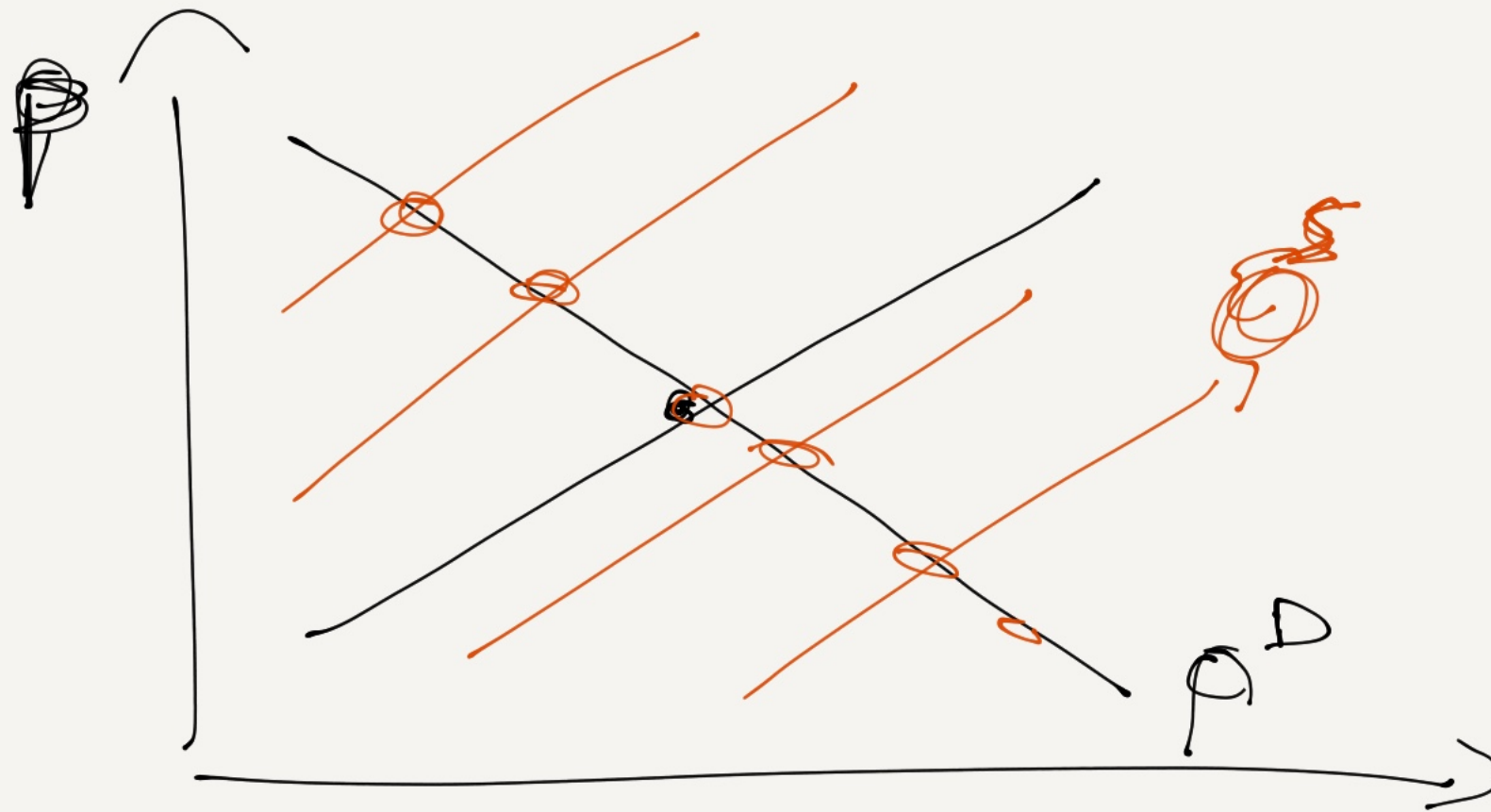
IDENTIFICATION



$\{Q_t, P_t\}$

$$\begin{cases} P_t^S = \alpha + \beta P_t + \varepsilon_t^S \\ P_t^D = -\beta P_t + \varepsilon_t^D \end{cases}$$

No Demand shock $\varepsilon_t^D = 0$ (Identifying assumption)



↳ β can be shared